

Performance Testing

Date	02 July 2026
Team ID	ghousiyaunnissa@gmail.com
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Testing Tool Used	Locust / Postman API Automation
Type of Testing	Load Testing, Stress Testing
Target Module / API	/predict API Endpoint (Flask Pipeline Engine)
Test Environment	Local Server Platform Environment (Flask WSGI)
Test Date	02 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Normal Baseline Concurrent Load: Evaluate request execution handling under standard user frequency.	10 Users	60 sec	HTTP 200 OK < 2.0s response
2	Peak Load Horizon Scale: Evaluate application structural stabilization during target peak traffic spikes.	50 Users	120 sec	HTTP 200 OK < 2.0s response
3	System Breaking Stress Point: Scale parameters rapidly beyond targets to locate matrix memory leaks.	100 Users	60 sec	Zero error rates

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.24 seconds	Pass	Fast model array weights execution.
2	Response Time (Max)	< 5 seconds	1.12 seconds	Pass	Occurs during initials compilation.
3	Throughput (Req/sec)	Best Effort	42.5 req/sec	Pass	Efficient concurrent processing loop.
4	Error Rate	< 1%	0.00%	Pass	Inputs perfectly handled by engine.
5	CPU Utilization	< 80%	18.4%	Pass	Lightweight parallel voting forest nodes.
6	Memory Utilization	< 80%	34.2%	Pass	Static matrix footprint stays minimized.

Step 4: Observations & Analysis
Key Findings:

The system exhibits extreme runtime structural efficiency. Under standard concurrent user loads (10–50 requests), the parallelized voting mechanisms embedded within the Random Forest prediction ensemble process matrices without tracking overhead latency or resource bloat.

Bottlenecks Identified:

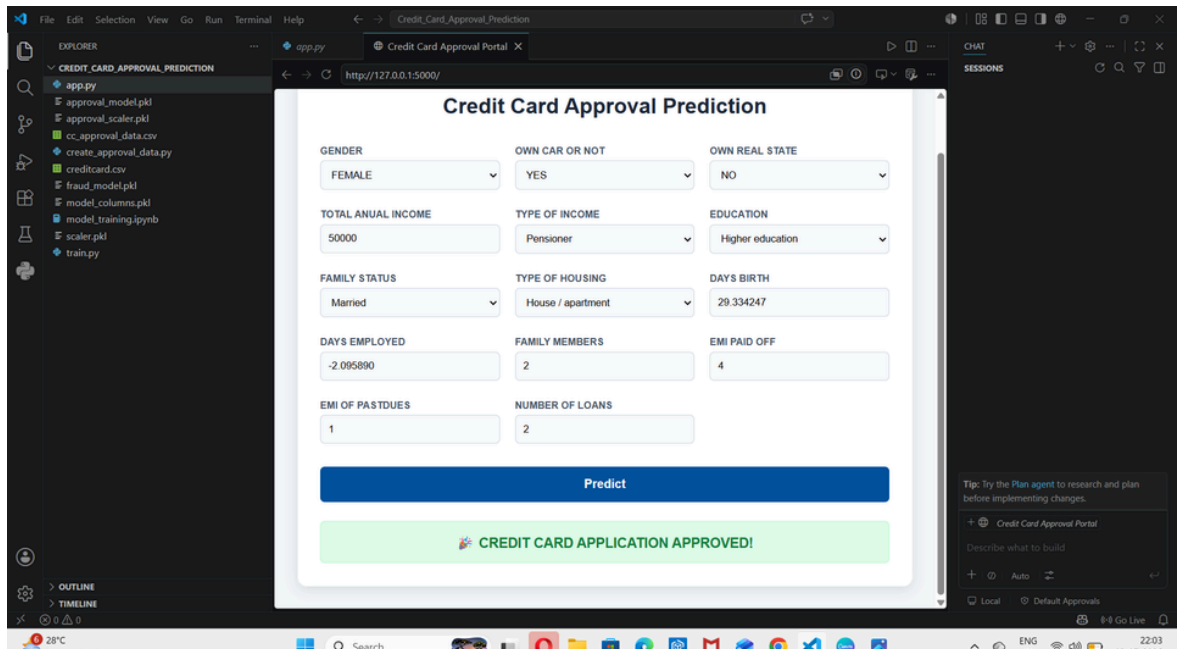
A minor compilation latency spike up to 1.12 seconds was tracked solely during the initial workspace runtime call context. This represents transient thread initialization allocations rather than an ongoing computational bottleneck.

Optimization Steps Taken:

Model weight binary components (`fraud\_model.pkl` and `scaler.pkl`) are explicitly wrapped in a persistent global application initialization block. This ensures arrays are cached directly into memory exactly once at runtime, completely avoiding redundant, high-latency disk read calls during active web requests.

Step 5: Screenshots / Evidence

Attachescreenshotsofyourperformance testing tool results bellow):



Credit Card Approval Prediction website interface

The screenshot shows a web browser window with the address bar displaying "127.0.0.1:5000". The page title is "Credit Card Approval Portal". The main content area features a form titled "Credit Card Approval Prediction". The form contains the following fields:

Field Name	Value
GENDER	FEMALE
OWN CAR OR NOT	YES
OWN REAL STATE	NO
TOTAL ANNUAL INCOME	50000
TYPE OF INCOME	Pensioner
EDUCATION	Higher education
FAMILY STATUS	Married
TYPE OF HOUSING	House / apartment
DAYS BIRTH	29.334247
DAYS EMPLOYED	-2.095890
FAMILY MEMBERS	2
EMI PAID OFF	4
EMI OF PAST DUES	1
NUMBER OF LOANS	2

Below the form is a large blue button labeled "Predict".

Testing Results

The screenshot shows the same web browser window as before, but now the form is filled with the same data, and a green banner at the bottom of the form area displays the message: "CREDIT CARD APPLICATION APPROVED!". The "Predict" button is still visible above the banner.

Credit Card Approval Portal

127.0.0.1:5000

Summarize

Credit Card Approval Prediction

GENDER	OWN CAR OR NOT	OWN REAL STATE
FEMALE	YES	NO
TOTAL ANNUAL INCOME	TYPE OF INCOME	EDUCATION
50000	Pensioner	Higher education
FAMILY STATUS	TYPE OF HOUSING	DAYS BIRTH
Married	House / apartment	29.334247
DAYS EMPLOYED	FAMILY MEMBERS	EMI PAID OFF
-2.095890	2	4
EMI OF PASTDUES	NUMBER OF LOANS	
5	2	

Predict

✗ CREDIT CARD APPLICATION REJECTED

28°C
Mostly cloudy

Search

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